# Concept Check: Introduction to Convolutional Neural Networks

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**1.** What is the primary difference between machine learning and deep learning? **1 / 1 point**

- ○ Machine learning is only used for numerical data and cannot be used for images
- ◉ Deep learning is a subset of machine learning well-suited to complex analyses
- ○ Machine learning automatically extracts trainable features

**Correct**
Many interacting computational layers allow deep learning models to tackle complex problems.

**2.** How do Convolutional Neural Networks extract useful information from images? **1 / 1 point**

- ○ By analyzing each pixel independently of neighboring pixels
- ○ By ignoring spatial information
- ◉ By performing convolution to learn and extract features

**Correct**
Convolution enables CNNs to capture patterns and details crucial for image understanding.

**3.** What is the purpose of activation layers in CNNs? **1 / 1 point**

- ○ To reduce the size of outputs
- ◉ To identify and forward important information
- ○ To learn complex relationships between features

**Correct**
Activation layers allow the network to capture essential features while suppressing irrelevant or misleading activations.

**4.** Which of the following describes the function of a pooling layer? **1 / 1 point**

- ◉ It reduces the size of outputs
- ○ It adds noise to the data
- ○ It performs convolution to learn and extract features

**Correct**
This reduction helps consolidate information from extracted features, letting the model extract more large scale and abstract features.

**5.** Why do CNNs repeat convolution, activation, and pooling layers many times? **1 / 1 point**

- ◉ To capture complex features
- ○ To eliminate the need for fully connected layers
- ○ To slow down the training process

**Correct**
As a general rule, networks with more convolutional layers will be able to extract more complex features.